

ABOUT ME

I am a meticulous individual with a strong background in Computational Science, data analysis, and data science. I have experience in data analysis, statistical modeling, and machine learning using Python, SQL, and various data analysis tools. I have a deep interest in data analysis, developing machine learning and artificial intelligence methods, data exploration, interactive data visualization, and optimizing ML algorithms using mathematical approaches. I am committed to generating reliable insights from complex datasets to support research and decision-making.

EDUCATION

•Bachelor of Science in Astronomy

Institut Teknologi Bandung, Indonesia

GPA: 3.25/4.0

- Start Date: 17 September 2020
- Graduation Date: 24 August 2024

•Master of Science in Computational Sciences

Institut Teknologi Bandung, Indonesia

GPA: 3.98/4.0

- Start Date: 9 September 2024
- Graduation Date: 12 August 2026

EXPERIENCE

•Graduate Researcher – P2MI

Institut Teknologi Bandung

Bandung, Indonesia

September 2024 – August 2026

- **Algorithm Development:** Designed and implemented GUMM, a custom Expectation-Maximization (EM)-based Gaussian-Uniform mixture model in Cython with OpenMP parallelization and an automatically-tuned membership threshold via spatial point-pattern analysis; open-sourced with full documentation (GitHub).
- **Statistical Pipeline Engineering:** Combined multiple unsupervised and probabilistic models (hierarchical density-based clustering, Xtreme Deconvolution Gaussian Mixture Model, Kernel Density Estimation – several with Bayesian formulations) into a single pipeline to extract and validate structured groups from a large-scale, high-dimensional dataset; first-authored peer-reviewed conference proceedings (arXiv:2604.23093).
- **Hybrid ML Pipeline (ENTRAP):** Built a hybrid pipeline combining density-based clustering with topological data analysis to recover edge cases missed by standard clustering, improving recall by 13–94% across 10 real-million scale datasets while maintaining precision at 1.00 and edge case recovery above 95% on 99 validation sets; documented as a proprietary package (GitHub).
- **Technical Communication:** Presented results at 2 conferences (1 international, 1 national).

Key Skills: Python, Cython, Bayesian Statistics, Unsupervised Machine Learning, Topological Data Analysis, Parallel Computing, Large-Scale Data Processing, Algorithm & Pipeline Design.

•Radio Signal Data Reduction & Analysis Collaborator – High-Energy Physics Project

Institut Teknologi Bandung

Bandung, Indonesia

September 2024 – November 2025

Data Pipeline Engineering: Used a signal-processing and data-reduction pipeline on a Linux HPC cluster to clean and extract low-signal data; Did advanced statistical analysis and fitting to extract physical parameters from cleaned data; results published in a reputable Q1 peer-reviewed journal.

Key Skills: Signal Processing, Data Reduction, Linux Server, Statistical Analysis, Python, Scientific Collaboration.

•Laboratory Assistant – Machine Learning & Advanced Machine Learning

Institut Teknologi Bandung

Bandung, Indonesia

February 2025 – June 2026

- Taught the fundamentals of supervised vs. unsupervised learning, data preprocessing, and model selection to graduate students.

- Designed hands-on classic NN, RF, Decision Tree, CNN, SVM, and K-Means assignments using TensorFlow, PyTorch, and Scikit-Learn.

Key Skills: Python, TensorFlow, PyTorch, Scikit-Learn, Statistical Analysis, Technical Mentorship.

•**International Research Internship – ASPIRE (Astrophysics Summer Program)**

Amsterdam, The Netherlands

University of Amsterdam

June 2026 – August 2026

- **Numerical Simulation:** Built a Python simulation pipeline from first principles on HPC infrastructure to test an alternative hypothesis for anomalous behavior in a complex dynamical system.
- **Time-Series Analysis:** Applied multi-start optimization, Box-Least Squares periodogram, and differential evolution for robust parameter estimation on synthetic time-series data; the pipeline was published on GitHub.

Key Skills: Python, Numerical Simulation, Time-Series Analysis, HPC, Scientific Computing.

SKILLS

Technical Skills

- Proficient in **Python**, **SQL**, and **LaTeX**, with a working knowledge of **C/C++**, used primarily for scientific data analysis, statistical modeling, and machine learning.
- Experience with **high-performance computing (HPC)** on Linux clusters, including **Cython**, **OpenMP**, **mpi4py**, and **mpich** for large-scale data processing and parallelization.
- Proficient in data science and ML libraries, including **Pandas**, **NumPy**, **SciPy**, **Scikit-Learn**, **TensorFlow**, **Keras**, **PyTorch**, etc.
- Experienced in **supervised and unsupervised machine learning**, including clustering, neural networks, and Bayesian statistical inference.
- Comfortable working across **Windows** and **Linux** operating systems.

Soft Skills

- Strong **problem-solving** and **critical thinking**, developed through independent, first-principles-driven research.
- Highly **adaptable and self-taught**, proven by transitioning into new technical domains without prior background.
- Effective **teamwork and collaboration**, built through multi-institution, international research projects.
- Strong **time management** skills, maintaining a 3.98/4.0 GPA while concurrently leading multiple research projects, mentoring responsibilities, and international activities & collaboration.
- Confident **presentation and communication** skills, demonstrated at multiple international and national conferences.

Languages

- **Indonesian:** Native
- **English:** High Intermediate

COURSES AND CERTIFICATION

•**Online Data Science Course**

Certification from DQ Lab – Focus on Data Exploration and Visualization

July - August 2021

- Exploratory Data Analysis with Python for Beginners.
- Data Visualization with Python (Matplotlib) for Beginners.
- COVID-19 Data Analysis in Indonesia using Python.
- Python for Data Professionals (Beginners).
- Data Science Training: Data Exploration and Visualization using Python.

•**Online Machine Learning Course**

Certification from freeCodeCamp – Focus on Machine Learning and Neural Networks

August 2024

- TensorFlow and Keras for Neural Networks.
- Gradient Descent Techniques and Data Preparation.
- Classification with Neural Networks.
- Natural Language Processing with Scikit-Learn and TensorFlow.

ACHIEVEMENTS

•**Dean's Award FMIPA ITB, Semester I - 2023/2024**

February 2024

Awarded to students with a GPA above 3.5 and at least 15 credit hours in the respective semester.

GPA: 4.00